BLG483E

AI Aided Computer Engineering

Midterm Exam

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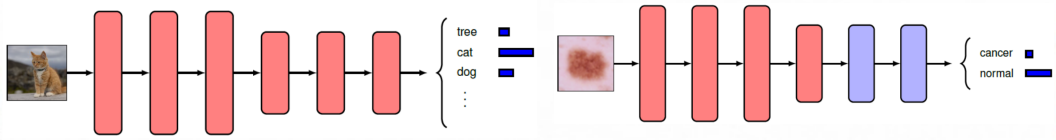
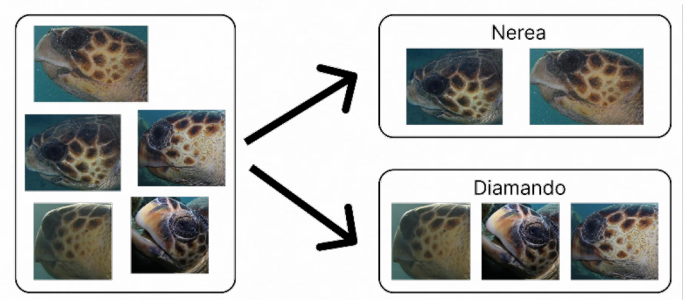
Q1) [15 pts] Freezing is a technique in transfer learning where some layers of a pre-trained neural network are kept fixed (their parameters are not updated) during training, while other layers remain trainable. For the given example below, only the last two layers of the network are trained. Why do we prefer freezing the earlier layers instead of the last layers? Explain.



Q2) You are given a collection of sea turtle head images captured underwater. Your documentary team has access to only two known turtles: Nerea and Diamondo. Your goal is designing a pipeline to identify new turtle head images. You are not allowed to train any new model. However, you are given access to CLIP, SAM and GPT-4o.



Q2-A) [5 pts] How SAM could be used in the pipeline? Why?

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Q2-B) [5 pts] Which model you would use to obtain the main representation vector in the model and why.

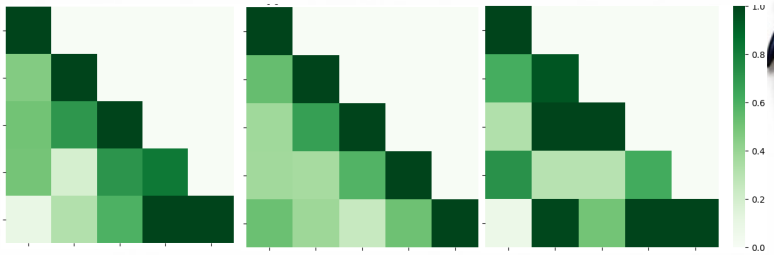
Q2-C) [5 pts] How you would compare the query image with the reference images of Nerea and Diamondo without any additional training.

Q2-D) [5 pts] How the final decision would be made when multiple reference images are available for each turtle.

Q3) [15 pts] For the given text tokens, using GPT-2, the attention maps given below are created. Match them!

	Token1	Token2	Token3	Token4	Token5
1	killed	sixteen	rebels	single	handed
2	the	bodies	hit	the	floor
3	blood	drops	on	the	snow

(A) (B) (C)



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Q4) Consider that you want to know whether penguins are mammals. You want a language model to answer this question using different prompting strategies.

Q4-A) [5 pts] Write a zero-shot prompt that helps the model correctly answer the question.

Q4-B) [5 pts] Write a few-shot prompt that helps the model correctly answer the question.

Q4-C) [10 pts] Modify your answer in **Q4-B** to represent CoT.

Q5) [10 pts] One of the major issues in GAN training is *mode collapse*. Explain what mode collapse is and why it occurs.

Q6) [20 pts] Fill the blank parts.

In GANs, the network responsible for generating fake data is called the _____.

In diffusion models, the forward process gradually adds _____ noise to the data.

The reverse process in diffusion models aims to _____ the noise step by step.

Stable Diffusion performs the diffusion process in _____ space instead of pixel space.

In diffusion models, the transformation from data to noise is modeled as a _____ chain.

The population in a genetic algorithm consists of multiple _____.

The quality of each individual is measured using a _____ function.

The process of choosing parents for reproduction is called _____.

The process of combining two parent solutions to generate offspring is called _____.

In EvoGym, the robot's control policy is typically learned using a _____.